

SANCHIT SAHAY

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RESEARCH INTERESTS

Software Supply-Chain Security, Software Systems, Operating Systems, Observability, Build Systems

EDUCATION

New York University, New York, NY

Ph.D., Computer Science

Sept 2026 – Present

Graduate School of Arts & Sciences

Advisor: Prof. Justin Cappos

M.S., Computer Science

Sept 2024 – May 2026

Tandon School of Engineering

GPA: 3.889 / 4

Manipal Institute of Technology, Manipal, India

B.Tech., Information Technology

June 2018 – July 2022

Minor: Big Data Analysis

GPA: 9.10 / 10

EMPLOYMENT

New York University, New York, NY

July 2026 – Aug 2026

Research Associate, Secure Systems Lab

- Extended and enhanced in-toto and go-witness's build observability attestations for SBOMit [1].

Commvault Systems, Bangalore, India

Jan 2022 – Aug 2024

Engineer, Virtual Server Agent Team

Python, .NET, VMware, Huawei Cloud

- Built data-protection software for VMware and Huawei private-cloud environments, deployed across 500+ enterprise and government organizations.
- Developed Commvault's VMware Cloud Director plugin, simplifying multi-step data-protection workflows for managed-service customers.
- Implemented agent-based Guest OS file recovery for VMware VMs, sending deduplicated data straight from source to storage and removing a bottleneck for workloads exceeding 10 GiB or 10,000 files.
- Extended Application-Aware Backup to VMware Cloud Director VMs, enabling granular backup and recovery of SQL Server and other applications.
- Refactored the Python SDK for VMware Cloud Director, improving end-to-end test reliability.

LegalAI, Remote

Apr 2021 – Dec 2021

Full-Stack & DevOps Intern

Node.js, React, Google Cloud Platform

- Designed and built a claims-processing platform of GCP-hosted microservices and React portals, handling legal-draft generation and review.
- Built the CI/CD pipeline around a custom App Engine-compatible local runtime, keeping local development in parity with GCP production.

RESEARCH

SBOMit (OpenSSF)

Sept 2025 – Present

OpenSSF initiative to augment Software Bills of Materials (SBOMs) with build time in-toto attestations.

- Extended witness, in-toto's Go implementation, with eBPF-based attestors that trace a build's filesystem and network activity at a low overhead to identify build inputs.
- Built Docker extensions for witness so builds that run in containers are still observed and attested.
- Reconciled in-toto attestation evidence with SBOMs produced by Software Composition Analysis tools, surfacing components that manifest-based SBOMs miss.
- Compared SBOMit's threat model and accuracy against Syft and Trivy, identifying build time attack vectors that manifest based SBOM tools cannot see [1].

Lind-Wasm & 3i (Secure Systems Lab)

July 2025 – July 2026

WebAssembly sandbox for isolating unmodified POSIX applications, with 3i routing system calls through composable userspace policies.

- Implemented and designed parts of policy composition on 3i so that independently written syscall policies (grates) can be chained, conditionally routed, or duplicated per process.
- Built an in-memory filesystem grate that gives sandboxed processes private, trusted file storage, used for C compilation inside Intel SGX enclaves.
- Built a Rust library of safe libc and 3i bindings, to extend Lind's support for Rust programs and grates.
- Built a benchmark suite measuring sandbox overhead, per-grate dispatch cost, and filesystem throughput, and used it to optimize syscall resolution.

PUBLICATIONS

1. **Sanchit Sahay**, Vyom Yadav, Abhishek Reddypalle, Uk Jang, Marco De Vincenzi, Santiago Torres-Arias, and Justin Cappos. 2026. *Over the Shoulder: Improving SBOM Accuracy by Watching the Build*. In Conference on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED '26), October 06, 2026, Prague, Czech Republic. <https://doi.org/10.1145/3848003.3848006>

INVITED TALKS

OpenSSF Community Day EU, Prague - Over the Shoulder: Improving SBOM Accuracy by Watching the Build. (Oct 2026)

PyCon US, Long Beach - Asleep at the Wheel: Getting your SBOMs to pay attention to Python builds. (May 2026)

KubeCon + CloudNativeCon, Atlanta - OpenSSF SBOMit Meeting. (Nov 2025)

OPEN SOURCE AND COURSE PROJECTS

Witness (CNCF)

github.com/in-toto/go-witness

Maintainer, witness and go-witness. Worked on integrating eBPF support for low-overhead build tracing.

HFS+ Port For FreeBSD: C

[FreeBSD Status Report](#)

Ported Apple's open-source HFS+ filesystem to FreeBSD 14, adapting kernel VFS-layer operations to modern FreeBSD interfaces. Developed userland utilities for mounting and management of HFS+ volumes.

Cargo: Rust

github.com/stupendoussuperpowers/cargo

Contributor to Cargo, Rust's official package manager and build system.

Improving Learned Bloom Filters

github.com/stupendoussuperpowers/wise-bloom-filters

Benchmarked optimization techniques for Learned Bloom Filters, evaluating the trade-off between false-positive rate and memory efficiency across Projection Hashing, Caching, and LoRA.

Talk2Data: Python, Google Cloud Platform (GCP)github.com/Sitanshuk/Talk2Doc

Built GCP data pipelines that extract and organize Gmail and Notion data into job applications, course materials, and deadlines. Implemented RAG-backed chat interfaces over that data using personalized LLMs, with load-efficient hosting on GCP.

MTA Ridership Prediction: Pythongithub.com/stupendoussuperpowers/mta-ridership

Trained ML models to predict NYC subway ridership from temporal and fare-class features; applied K-Shape clustering to uncover neighborhood-level ridership patterns across the system.

TEACHING

Course Assistant, NYU CS-GY 9223: Software Supply Chain Security (Fall 2026)

Teaching Mentor, Manipal Institute of Technology: Problem Solving Using Computers (2019)

TECHNICAL SKILLS

Languages: Rust, Python, Go, C/C++, Node.js, Java, C#

Frameworks: eBPF, React, Next.js, .NET, Android SDK

Databases: MongoDB, SQL Server, PostgreSQL, Redis

Cloud & Virtualization: GCP, AWS, VMware vCenter and Cloud Director, Huawei FusionCompute, Docker, Kubernetes

ML/Big Data: Pandas, NumPy, Keras, Spark, Hadoop, neo4j, CUDA